# MVP CT Flight-Test Matrix - Release Candidate 1

Radiation Shielding Assessment App - CT MVP release-gate record

| Release-gate item | Recorded value |
|---|---|
| **Overall MVP CT matrix** | PASS - CT-01 to CT-20 |
| **Full pytest regression** | PASS - 251 tests passed |
| **Commit hash** | 8a18f9e |
| **Branch** | dev |
| **Date** | 15 Jun 2026 |
| **Tester** | SBH |
| **Release-candidate label** | MVP CT RC1 baseline |

## Release-gate interpretation

This matrix records the CT MVP release-candidate verification baseline. The release gate focuses on core confidence: source identity and ksec, geometry distances, barrier calculations, PASS/REVIEW/FAIL status consistency, source-readiness gating, save/reopen persistence, and PDF/CSV/JSON agreement with the UI.

**Remaining CT MVP tests:** none recorded. Optional post-MVP polish remains non-blocking unless it affects source identity, calculation integrity, status consistency, or report/export correctness.

## Flight-test matrix - part 1

| ID | Test / gate | Status | Evidence / notes |
|---|---|---|---|
| Step 1 | Pytest baseline before matrix | **PASS** | Baseline test suite passed before manual CT flight testing. |
| CT-01 | Clean database scanner | **PASS** | Revolution Apex; Geometry ksec displayed; Sources basis Database; Results, PDF, CSV and JSON generated. |
| CT-02 | NULL ksec scanner with no override | **PASS** | Brilliance 64; NULL/manual override required; source readiness blocking; warning dialog; draft PDF only; no CSV/JSON. |
| CT-03 | NULL ksec with complete manual override | **PASS** | Calculations run; manual override shown as REVIEW not Database; audit appears in report/source basis; CSV/JSON generated if no other blockers. |
| CT-04 | Manual override with incomplete audit | **PASS** | Manual override shown as REVIEW; audit incomplete warning; review/not final certification wording; certification not permitted. |
| CT-05 | Move scanner then export without visiting Results | **PASS** | Export accepted after scanner movement. Note retained: room diagram in PDF would further confirm isocentre location. |
| CT-06 | Change room dimensions then export | **PASS** | Room dimensions 7.0 m / 6.0 m; barrier distances updated correctly; Results and physics reflected new geometry. |
| CT-07 | Move source position and verify distances | **PASS** | Room 7.0 m x 5.0 m; source x=3.0, y=5.0; distances N=2, S=5, E=2, W=3 m. |
| CT-08 | Floor/ceiling distance behaviour | **PASS** | Room height 4.0 m; Floor distance 1.0 m; Ceiling distance 3.0 m; Results updated. |
| CT-09 | Change wall shielding and update results/export | **PASS** | Wall shielding change propagated to Results and export path. |

## Flight-test matrix - part 2

| ID | Test / gate | Status | Evidence / notes |
|---|---|---|---|
| CT-10 | Add door opening | **PASS** | Door workflow passed. |
| CT-11 | Viewing panel/opening inheritance | **PASS** | Opening row shown; parent wall linkage and inheritance accepted; export path passed. |
| CT-12 | Door/opening after moving geometry | **PASS** | Door on E wall; geometry changed; parent and opening distance updated; PDF/CSV/JSON included updated opening. |
| CT-13 | Clean Draft PASS wording | **PASS** | Clean draft PASS wording accepted. |
| CT-14 | FAIL/REVIEW report wording and consistency | **PASS** | Barriers tab, Results tab, Geometry overlay, PDF and CSV/JSON payload consistent; borderline low-margin barriers promoted to REVIEW. |
| CT-15 | Incomplete NULL-ksec source gate | **PASS** | Brilliance 64 with no override; source readiness blocking; warning dialog; draft PDF incomplete/not-for-certification; no CSV/JSON; certification not permitted. |
| CT-16 | Complete manual override review export | **PASS** | Brilliance 64 manual ksec 0.015 with complete audit; PDF and CSV/JSON generated; manual override wording; certification not permitted. |
| CT-17 | Save/reopen database scanner | **PASS** | Database scanner save/reopen path passed. |
| CT-18 | Save/reopen manual override | **PASS** | Manual ksec retained in JSON; Sources retained 0.015; Geometry database status NULL/manual override required; source basis preserved. |
| CT-19 | Full clean regression after source-identity fixes | **PASS** | Revolution Apex; Geometry label, Sources ksec, Barriers/Results, overlay and PDF all consistent; no source-readiness warning. |
| CT-20 | Save/reopen/export after source-identity fixes | **PASS** | Revolution Apex saved and reopened; Geometry/Sources/Barriers/Results/overlay/PDF/CSV/JSON all consistent; no warning. |
| Full regression | Complete pytest suite | **PASS** | 251 tests passed at commit 8a18f9e on branch dev, 15 Jun 2026, tester SBH. |

## Core confidence gates verified

| Gate | Evidence |
|---|---|
| Source identity and ksec consistency | Default/startup, dropdown changes, save/reopen and output generation verified through CT-14F/CT-19/CT-20. Sources tab owns scanner identity and ksec; Geometry displays the same identity. |
| Geometry and barrier distance propagation | Room dimension, source-position and vertical barrier distance changes verified through CT-06, CT-07 and CT-08. |
| Opening inheritance and export | Door and viewing-panel behaviour verified through CT-10, CT-11 and CT-12. |
| Manual ksec workflow | NULL ksec blocking, complete manual override, incomplete audit and save/reopen persistence verified through CT-02, CT-03, CT-04, CT-15, CT-16 and CT-18. |
| UI/report/export status consistency | Barriers tab, Results tab, Geometry overlay, PDF, CSV and JSON use consistent user-facing PASS/REVIEW/FAIL status after CT-14 fixes. |
| Regression test suite | Full pytest regression passed: 251 tests at commit 8a18f9e. |

## Remaining / deferred items

| Item | Status |
|---|---|
| CT MVP release-gate tests | None remaining - CT-01 to CT-20 PASS. |
| Room diagram in PDF report | Deferred polish noted from CT-05 unless required for release documentation. |
| UI/layout niceties | Non-blocking if they do not affect source identity, calculations, statuses or exports. |